Video Game Sales Analysis

Task 1 | Findings and business interpretation

Scope: 16,719 source records from the main file dated 22 December 2016 in the assigned Kaggle dataset. Sales measure millions of copies, not revenue or profit. Results below describe the supplied historical sample.

Measure	Result
Rows after cleaning	16,717
Recorded global sales	8,917.88 million copies
Leading title	Wii Sports - 82.53 million
Leading platform	PS2 - 1,255.64 million
Leading publisher	Nintendo - 1,788.81 million
Mean / median per release	0.533 / 0.17 million

Cleaning decisions

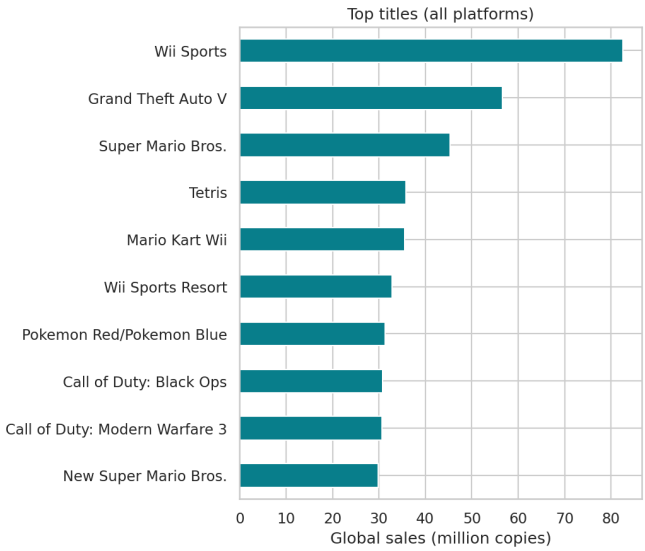
No exact duplicate rows were found. Two unnamed records were excluded (2.42 million recorded copies). Four years beyond the 2016 snapshot were set to missing; 273 records consequently have no usable release year. Unknown publishers and other categorical gaps were retained as Unknown; absent scores were not invented.

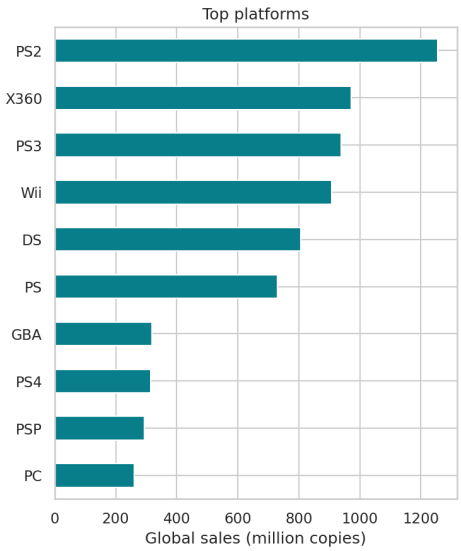
Two non-identical Madden NFL 13 / PS3 / 2012 records share a release key. They remain flagged rather than arbitrarily removed. Keeping only the larger entry would reduce the global total by 0.01 million copies. Regional and global sums can differ slightly because fields are rounded.

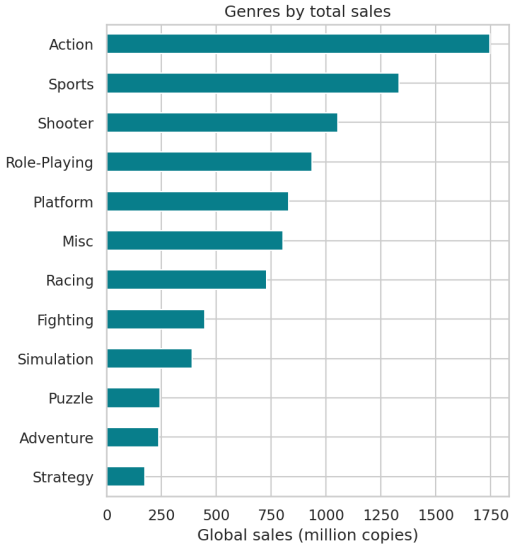
Interpretation boundary

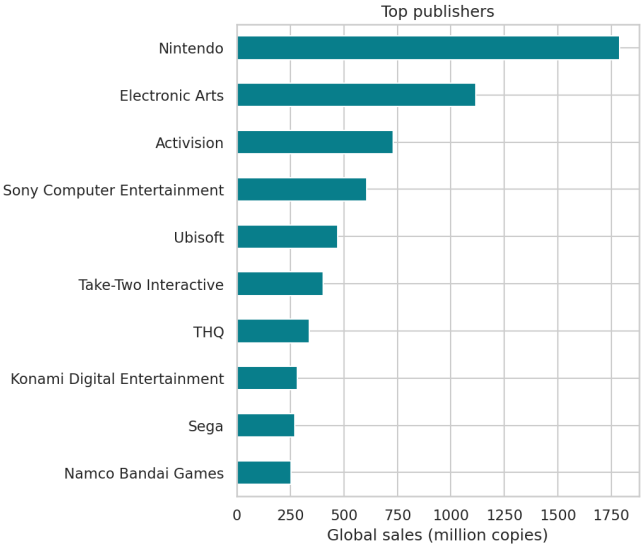
A platform-release record is not a unique game. Title totals combine matching names across platforms, which can also combine same-name editions. Coverage and estimation methods limit market-wide generalization. The data do not establish current market share or profitability.

Leaders and catalog scale









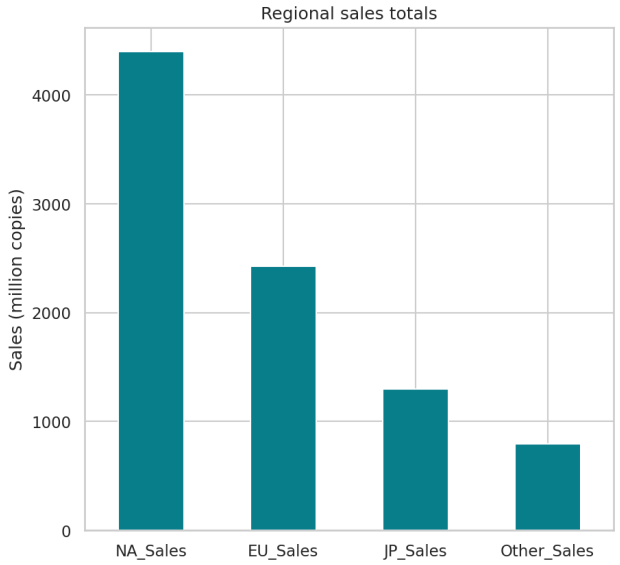
Key findings

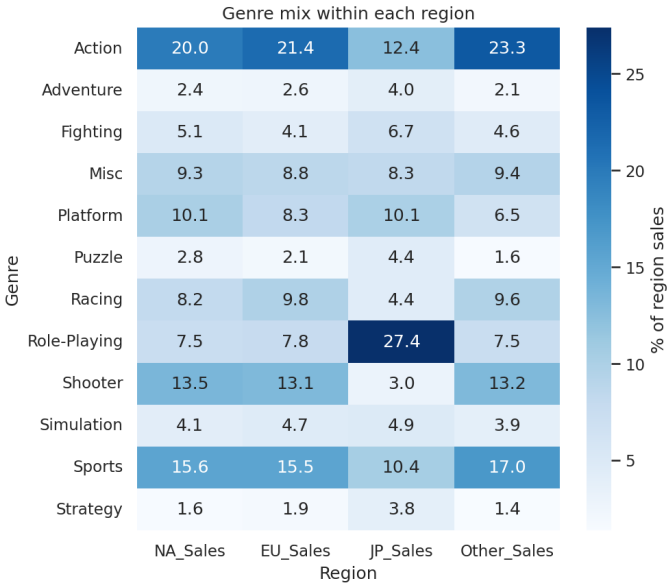
Action leads total genre sales at 1,745.27 million copies, while Platform has the highest median per release at 0.27 million. A high aggregate total therefore does not imply that the typical release has the best outcome.

The top ten aggregated title names contribute 4.61% of recorded global sales. Mean release sales (0.533 million) exceed the median (0.17 million), consistent with a right-skewed distribution containing large hits.

Practical implication: compare typical sales, catalog size and release exposure together before selecting a genre or platform benchmark. Profitability requires prices, costs and margins beyond this dataset.

Regions, timing and conclusions





North America represents 49.4% of the sum of the four regional sales fields. Japan's top genre is Role-Playing, showing why regional mix matters when using these historical benchmarks.

The 2008 release cohort has the highest recorded global total: 671.79 million copies. This is cumulative sales grouped by release year, not sales made during 2008. Newer cohorts have shorter exposure; the 2016 snapshot year is incomplete.

Recommended next steps

Use region-specific comparisons and median sales alongside totals. Validate the flagged release-key duplication and missing years. Before making investment decisions, obtain newer and more complete market coverage plus development, marketing, pricing and distribution data.

Reproducibility

Analysis.ipynb contains the executed cleaning log, descriptive statistics, comparisons, five saved figures and validation assertions. The included CSV is the exact input to this report. Re-running against a different file can change all findings.

Source: Kaggle - Video Games Sales Dataset
Main file: Video_Games_Sales_as_at_22_Dec_2016.csv. Retrieved 5 November 2026.